



Atomic Engines For Aircraft

Supersonic bombers capable of flying non-stop around the world seem definitely on the way.

By ANDREW KALITINSKY

Cutaway Drawings by Douglas Rolfe

ATOMIC energy promises to power airplanes combining extremely high speed and almost unlimited range. The application of nuclear energy may eventually make possible a supersonic bomber capable of flying around the world without pausing to refuel.

We have not manufactured an atomic power plant for aircraft, but we can say now that the practical development of atomic energy as a source of power calls not so much for new discoveries as for a great deal of engineering work. The practical use of atomic power can become a reality if physicists, engineers and industry take an increasingly active part in its development.

There are definite limitations on the possible performance of chemically fueled aircraft, since the required gross weight of the plane increases with both range and speed. For a given speed the required gross weight increases rapidly with range as the amount of required fuel increases. If we try to design an airplane of greater range at this given speed, we cannot do it no matter how large a plane we select. As the airplane speed is increased, the gross weight for any selected range rises. Even more important, the maximum range that can be achieved decreases as the airplane's speed increases.

Because of this limitation on the speed-range performance of chemically fueled planes (this includes jets), we must find a new fuel. Our present supersonic airplanes run out of fuel in a matter of a few minutes. With atomic energy, they could keep on going, for the aircraft's fuel supply would remain nearly constant.

Several basic types of power plants can be adapted to utilize atomic power for the propulsion of aircraft. They are all thermal power plants, since fission energy is released mostly in the form of heat.

One example of a thermal power plant which could be used with atomic fuel is the *closed-cycle turbine*. It may be a steam or mercury turbine. Steam is generated in the nuclear reactor by the heat released there. The reactor therefore replaces the boiler of a conventional power plant. The steam expands through the turbine, which drives the propeller. The steam is then condensed in an air-cooled condenser and is forced back into the boiler-reactor by a feed pump. This type of power plant requires the use of both a propeller and an air-cooled condenser, which will evidently limit the airplane speed at which it can be used. For that reason it does not appear as practical as other possible applications.

Another application of atomic energy is in the *turbo-jet*.

What is done here is simply to remove the combustion chamber of the conventional turbo-jet and replace it with a nuclear reactor. Air is compressed and forced through the reactor, where it is heated by convection instead of by combustion of fuel. It then expands partially in the turbine, enough to provide sufficient power to drive the compressor, and finally in the jet nozzle where it creates propulsive thrust. This type of power plant is very well suited to high-speed airplanes.

On paper, the *ram-jet* is the simplest type of power plant conceivable. Air enters the diffuser at the front end of the engine and is compressed by the forward speed of the plane. It then passes through the nuclear reactor where it is heated to a high temperature, and goes into the exhaust nozzle, where it expands and acquires a high velocity to provide thrust. The ram-jet requires a high flight speed to function at all and becomes really effective

at extreme speeds high in the supersonic region. The air temperatures required by the ram-jet are very high, considerably higher than those needed by the turbo-jet. At the same time, the ram-jet is very sensitive to pressure drops caused by the internal flow resistance of the reactor or combustion chamber. Good heat transfer conditions must always be paid for by appreciable pressure drops. The ram-jet is therefore not quite the simple problem it appears to be at first glance.

In the application of nuclear energy to a *rocket*, a propellant (liquid hydrogen for example) is pumped out of the tank and through the reactor, where it is vaporized and heated to a high temperature. It then escapes at high velocity through the exhaust nozzle. The rocket is driven by the recoil of the escaping propellant and thus is not dependent on atmospheric air

for its functioning. It can, therefore, operate outside the earth's atmosphere.

One may well ask where the advantage lies in using nuclear energy for rockets, since their endurance is limited and they can operate only until the fuel is exhausted regardless of the practically unlimited supply of energy in the reactor. Nuclear energy offers a definite advantage in rockets because the maximum impulse, the pounds of thrust that can be obtained from each pound of fuel used per second, results from a combination of the highest temperature and the lowest molecular weight of fuel.

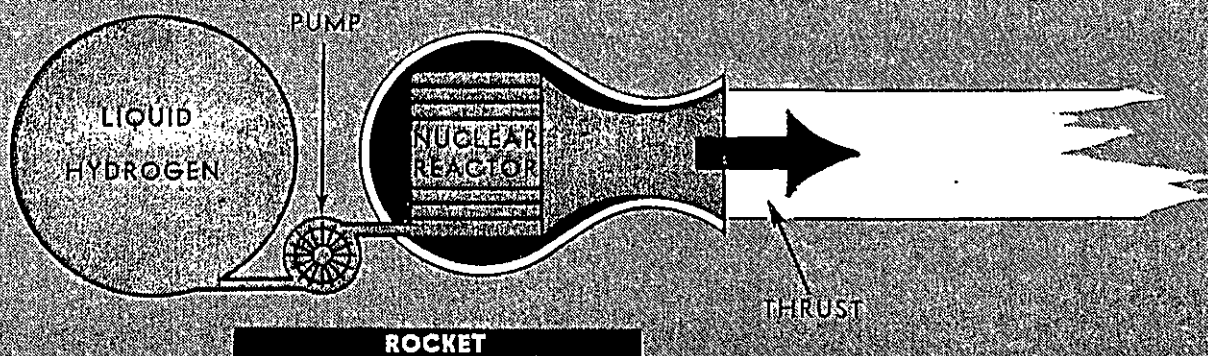
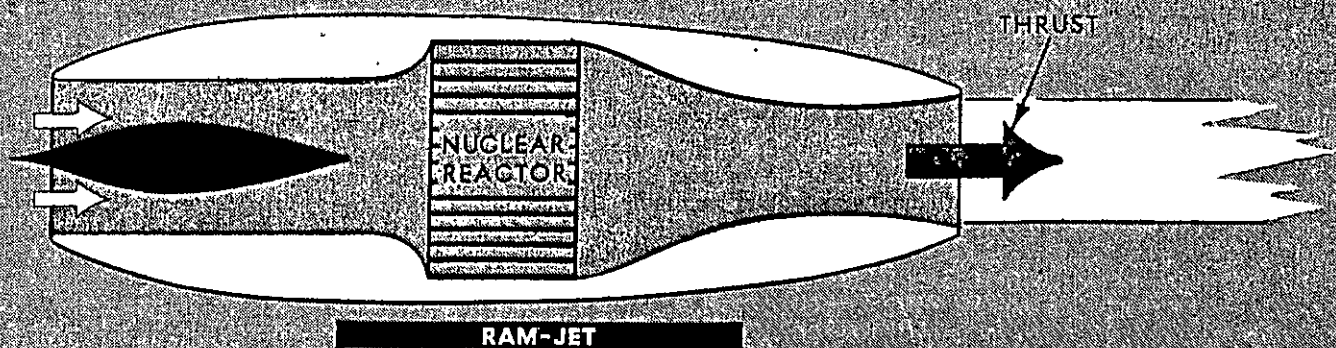
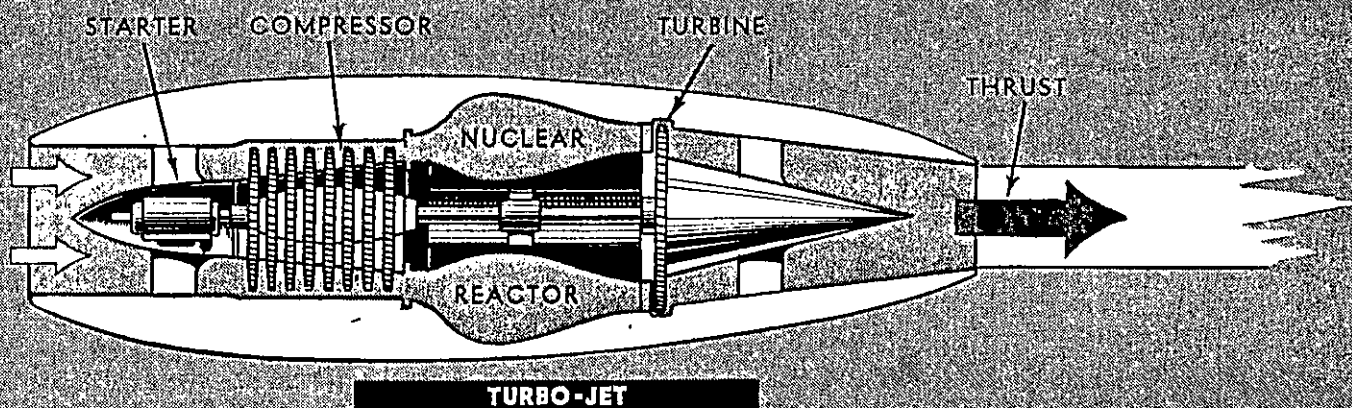
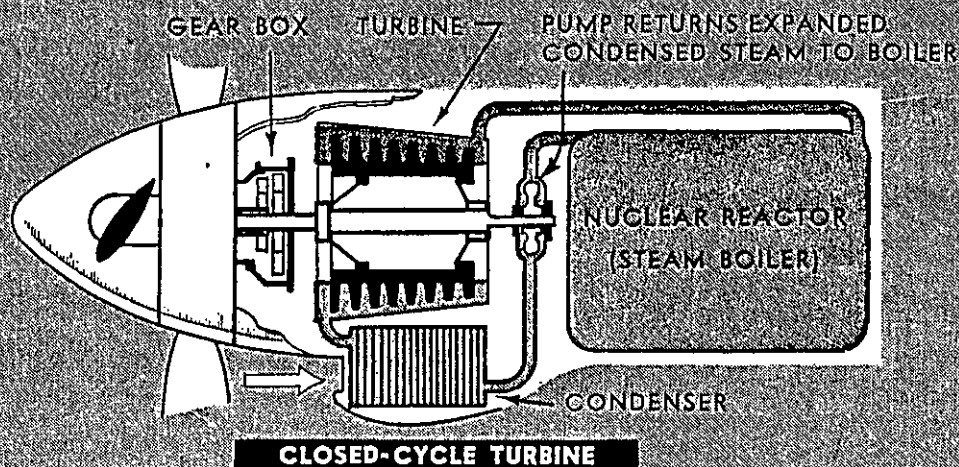
The high temperature is obtained normally in a chemical rocket by the combustion of a fuel and an oxidizer whose products of combustion (Continued on page 53)



Atomic Physicist

Andrew Kalitinsky, at 34, is chief engineer of Fairchild's NEPA (Nuclear Energy for the Propulsion of Aircraft) Division. This article is reprinted from *The Pegasus*, a magazine published by Fairchild.

FOUR POTENTIAL TYPES OF ATOMIC ENGINES



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Atomic Engines

(Continued from page 22)

are then used as the propellant. Since the propellant is the result of the combination of at least two atoms, its molecular weight will be fairly high. If hydrogen and oxygen are used, the resulting propellant is water vapor with a molecular weight of 18. If nuclear energy is used to provide the higher temperature, there is no need for the process of combustion and a very light propellant, like hydrogen, with a molecular weight of 2 can be used. The specific impulse of pure hydrogen at the same temperature is three times that of water vapor.

Many of the problems which lie in the path of the practical realization of atomic aircraft power plants are connected with the use of high temperatures. Unlike chemically fueled engines, high thermal efficiency is not the primary reason for wanting high temperatures in a nuclear engine. Specific fuel consumption is of vital importance in a chemically fueled engine. However, for a fuel with a heating value of 40 billion BTU per pound (like uranium), specific fuel consumption is vanishingly small, from a performance standpoint, and thermal efficiency in itself is not a primary objective. High performance is, however, and high performance calls for high operating temperatures. For instance, a high thrust per pound of airflow per second is required in turbo-jets for high-speed airplanes. This requires a high energy input for each pound of air and, therefore, a high air temperature.

The temperature problem is aggravated by the heat-transfer problem. We actually are taking a long step backwards along the road of development in the light-weight engine. The great advance in light-weight design, which made the automobile and airplane possible, came to pass when it was learned how to avoid transferring heat through walls, as in a boiler. In other words, with the advent of internal combustion.

In a nuclear engine, the heat is generated in the solid portions of the reactor and must be transferred to the working fluid through heat-transfer surfaces. Fortunately, we have learned a lot about heat transfer in the past 50 years so we are not starting quite as far back as the men who did the initial development work on the gasoline engine.

One basic fact remains, however, and that is that the internal surfaces of the reactor must be at a temperature higher than the highest temperature of the working fluid, since heat must flow from the reactor into the fluid. It can only flow if there is a positive temperature difference. This is very different from the internal combustion engine where, as in the automobile, the cycle temperatures reach 4,000° while the internal surface temperatures can be kept down to around 500° by external cooling. On the other hand there are no moving parts in the reactor and the reactor elements are not subjected to the high dynamic stresses which are encountered in such parts as pistons, valves or turbine buckets. It can

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be said broadly that the temperatures necessary for the realization of an atomic engine are not unreasonable from the metallurgical standpoint, but they do nevertheless engender a number of difficult problems.

One of these problems is the protection of the uranium in the reactor against corrosion by the working fluid and, conversely, the prevention of the escape of the radioactive fission products from the reactor into the working fluid. This is a problem of diffusion and diffusion rates generally go up with increasing temperature. The development of adequate canning for the high-temperature reactor elements is, therefore, a major problem.

Another crucial problem, as in any aircraft, is weight. In an atomic engine, it is necessary to use large amounts of mass to stop radiations emitted during the fission process. A comparison with a conventional airplane can help us visualize the magnitude of the weights which may be allotted to shielding. The weight of the propulsion system comprises, substantially, three major items: the power plant proper, the fuel and the fuel tanks. In the nuclear aircraft, these correspond to the "engine" (compressor-turbine assembly in a turbo-jet), the nuclear reactor and the shielding.

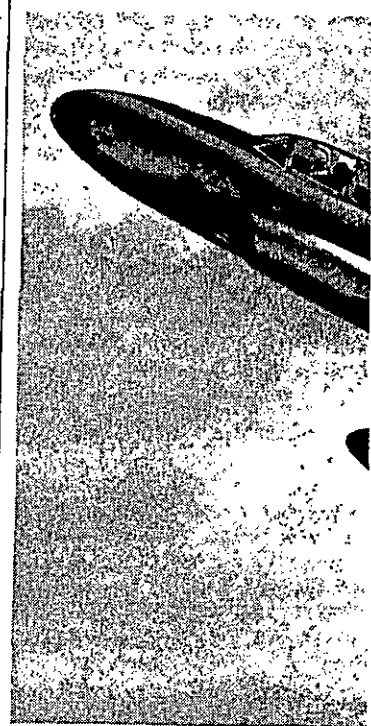
The "engine" component of the nuclear installation may be expected to weigh approximately the same as a conventional power plant of the same horsepower or thrust rating. The weight of reactor and shielding is therefore equivalent to the weight of the fuel and fuel tanks in a chemically fueled aircraft.

The percentage of the airplane gross weight which can be allotted to fuel and fuel tanks, or to reactor and shielding, depends primarily on the structural refinement of the airplane. In present-day long-range airplanes, it represents a very high percentage of the gross weight and is, in some cases, well in excess of 50 tons. It is therefore obvious that the problem of an atomic-powered airplane is in an entirely different category from the problem of an atomic-powered automobile, although they have often been mentioned together in discussions of atomic energy.

Not only the absolute weight of the shielded reactor is important, but also its specific weight, or its weight per unit of thrust produced by the power plant. The weight of the shielded reactor is tied in with the gross-weight of the airplane by structural efficiency. The aerodynamic efficiency of the plane, the ratio of lift to drag, ties in the required power plant thrust with the airplane gross weight. The required lift is equal to the gross weight, while the resultant drag must be overcome by the power plant thrust. By combining these relationships, we arrive at a formula which ties in the power plant performance with the aerodynamic and structural characteristics of the airplane; the specific weight of the reactor must be less than, or equal to, the lift-to-drag ratio times the ratio of permissible reactor weight to airplane gross weight. The range of an atomic-powered airplane is practically unlimited and is not a design variable.

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—Harold Helfer

Shielding weight is influenced about equally by two entirely separate variables. One is the size of the reactor core around which the shielding must be wrapped. The other is the thickness of the shielding itself or, more precisely, the mass thickness needed to stop gamma rays and neutrons emitted by the fission process.

Making the reactor small is an obvious way of reducing weight of shielding. However, it means releasing energy at a high rate in a small volume. Although a nuclear reactor can potentially generate heat at a practically unlimited rate, this heat must be conducted from the interior of the fuel rod to its surface. It must then be transferred from the surface to the working fluid, meaning high temperature differentials, high pressure drops, large internal surface areas and all other difficulties associated with high power densities. Furthermore, the smaller a reactor is, the larger the ratio of its surface area to its volume and the easier it is for

neutrons to escape without causing new fissions. A smaller reactor may therefore require a larger investment of uranium in order to sustain the chain reaction and this may prove undesirable from an economic standpoint.

As far as mass-thickness of the shield is concerned, the best materials for stopping a given type of radiation of a given energy are relatively well known. However, the radiations emitted by a nuclear reactor are of two major types, neutrons and gamma rays, each covering a wide spectrum of energies. As they pass through the shield, these radiations gradually change their character. The fast neutrons slow down by successive collisions with nuclei of the shield material; the gamma rays are absorbed and in turn softer X-rays are emitted. A material which is good for stopping gamma rays may not be the best when gamma rays and neutrons are considered together, and a material which is good for the innermost part of the shield may not be the

best for the outer layers. There is, therefore, considerable room for weight reduction by ingenious design.

Since we are dealing with a relatively thick shield surrounding a relatively small reactor, the apparent paradox holds true that, for a given mass thickness, the greater the density of the shielding material the lighter the shield. If the density of the shielding material is doubled and the thickness of the shield is cut in half, the volume of the shield is decreased by considerably more than a factor of two and the total weight is reduced accordingly.

From the point of view of conventional power plant experience, shielding may be used as a good example of some of the unusual aspects of nuclear engineering. If two inches of lead reduce some radiations to one-tenth their original intensity, four inches of lead will reduce them to one-hundredth, and six inches to one-thousandth of the original intensity. The radiation intensities encountered in nuclear reactors must be reduced by factors of many billions before they are safe for humans.

A good reflector may turn back as many as 90 per cent of the neutrons escaping from the reactor. It would be wrong, however, to assume that this would solve nine-tenths of the neutron shielding problem. It is more likely to represent only about 10 per cent of the job. In addition, it is very important to prevent the escape of seemingly small amounts of radiation through joints or seals in the shield around the reactor.

The choice of materials for shielding also presents unusual problems. Existing reactors mostly use large quantities of concrete for shielding, but concrete is hardly a suitable aircraft material. The effect of intense radiation on the properties of the materials must be considered too. It has been found that the electrical resistance, the elasticity and the heat conductivity of graphite all change with exposure to intense neutron radiation.

The atomic powered airplane must be designed for very high speeds to take advantage of the special characteristics of atomic power. It certainly will be a large airplane. It will have to be designed for a landing weight which is equal to take-off weight, since practically no fuel will be used up in flight. Because of radiation, the crew locations should be placed as far away from the power plant as possible. That would permit some weight saving in the shield.

Structural requirements will be somewhat different from those of conventional airplanes, since the fuel load will be concentrated in one spot in the reactor, rather than widely distributed. In this respect, however, the atomic airplane may not differ much from some of the thin-winged, high-speed airplanes now under development which do not use wing tanks.

These are some of the problems associated with the development of a nuclear aircraft power plant. They are not insurmountable problems, but such a development is not by any means an easy one.

END



MRS. AMERICA IS A PILOT

MR. AMERICA—beautiful 21-year-old Maria Strohmeir—has kept pace with the air age since she was 17. As a teen-age model in Philadelphia, she spent her earnings on flight lessons. Today, she and her husband have one major ambition: to fly around the U.S. in their own airplane.

When she expressed the desire recently on a radio show, Universal Pictures and Aeronca Aircraft Corporation promptly gave the young couple the opportunity. They spent several weeks touring the nation in a new Aeronca Sedan, promoting Universal's forthcoming movie, "You Gotta Be Happy."

Incidentally, Maria's romance and marriage began at an airport, where she met her husband. They honeymooned in an Aeronca Champion, and now both the Strohmeirs (they hold private licenses) are trying to wangle a chance to make a parachute jump!

More interesting than those of the planes she flies, Maria's specifications read: Height 5'6", weight 120, bust 35, waist 25, hips 36, thigh 20, ankle 8.



A regular FLYING reader, Maria has saved copies of the magazine for the past four or five years.

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